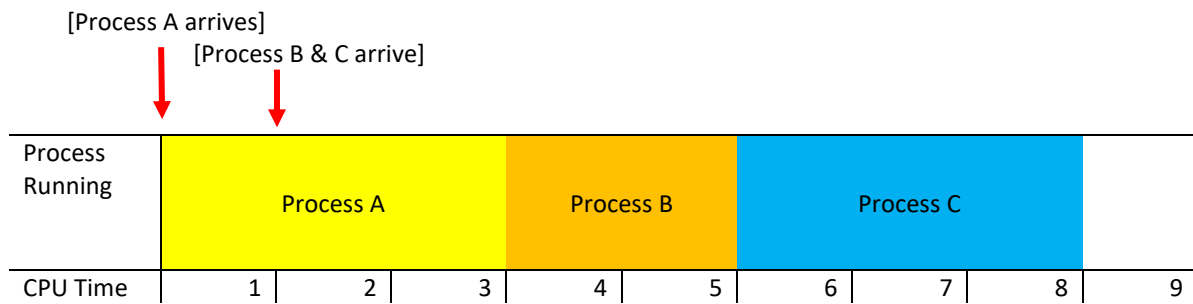


Quiz 2 – Applied OS: Cyber Defense & AI Agents

L1: Student Name: _____

Q1				Q2	Total
(2)	(2)	(2)	(2)	(2)	(10)

Q1 (8)



a) What is the average turnaround time? (2)

Turnaround Time (TT) = Completion Time - Arrival Time.

- Process A $TT_A = 3 - 0 = 3$
- Process C $TT_C = 5 - 1 = 4$
- Process B $TT_B = 8 - 1 = 7$
- Total $TT = 3 + 4 + 7 = 14$
- $TT_{ave.} = \frac{14}{3} = 4\frac{2}{3} = 4.666$ time units.

(b) What is the average response time? (2)

Response Time (RT) is calculated as First Run Time - Arrival Time.

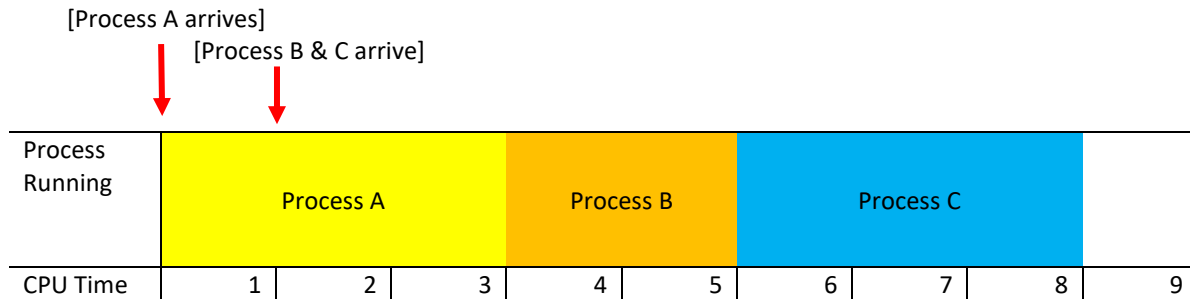
- Process A: $RT_A: 0 - 0 = 0$
- Process C: $RT_C: 3 - 1 = 2$
- Process B: $RT_B: 5 - 1 = 4$
- Total $RT = 0 + 2 + 4 = 6$
- $RT_{ave.} = \frac{6}{3} = 2$ time units.

(c) What type of scheduling policy is this? (2)

FIFO, SJF, and STCF are all correct.

- (d) Using a drawing, show how the STCF (shortest time to complete first) policy changes the way the process runs in the CPU. (2)

It will remain the same:



- Q2 The wait function in lab 02 was run in the following way: (2)

```
int wc = wait(NULL);
```

Here, we are giving NULL as arguments to the function. In essence, it allows the parent to wait for the child. Now, can process A wait for an entirely different (unrelated) process B (one not created using fork, thus not having a parent-child relationship) using `wait()`?

No, Process A cannot use `wait()` or `waitpid()` to wait for an entirely unrelated Process B. The wait family of system calls in POSIX operating systems is strictly restricted to the parent-child hierarchy.